Models for the evolution of a phylogenetic tree

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1 Introduction

Phylogenetic models aim at reconstructing and analyzing the phylogenetic tree underlying a set of extant species (or other higher taxa). This tree is fundamentally evolved due to a series of speciations and extinctions of species, which makes it reasonable to model using the birth-death model or more generally by a Markov branching process (T. E. Harris 1963; Nee 2006). But comparisons with empirical trees have shown that these models rarely conform to the data, in particular when considering the balance of the tree, i.e. the relative size of subtrees (Blum and François 2006; Jones 2011b).

Branching processes have a rich history and have been extended significantly beyond the Markov assumption (T. E. Harris 1963; Jagers 1975). Hence a natural extension of the phylogenetic models, which are mostly assuming the Markov property, is to consider these non-Markovian branching processes, often called age-dependent or Bellman-Harris processes, as the stochastic process underlying the phylogenetic tree, and investigate whether such a model can do a better job at fitting the data. However, this has only been done by few in the phylogenetic community (Jones 2011a; Gernhard, Hartmann, and M. Steel 2008; Hagen et al. 2015), and traditionally branching processes are mostly considered by their size over time, and their asymptotic behavior, not the underlying branching structure.

In this report, we will review stochastic models for phylogenetic trees (and related structures), starting with simple probability measures on discrete tree spaces and going into more complicated models in which we take time and further biological forces into account. In particular, we will consider a model proposed by Jones (2011a) and Hagen et al. (2015), building upon Bellman-Harris branching processes. We begin with a review of some graph theory, as the theory requires quite a few notions about trees. We will also consider some combinatorial results about these trees. We will then define two basic models on the space of phylogenetic trees, and see that there exists a larger family of models, which includes them both, and which can produce trees of varying balance. Then we turn to evolution as happening through time by the Bellman-Harris process by Jones (2011a),

with a detour into the mathematical definition of a branching process, ending with a definition of the tree generated by a branching process. The last part of the report shows how to calculate several probabilistic quantities for the model, and most importantly a formula for the density of a tree generated under the model is shown. We see that the simple pure-birth model by Yule (1925) is a special case of the process.

2 Trees

Trees form the basis for phylogenetic thinking, so a mathematical theory of evolution requires a definition of a phylogenetic tree. To write down such a definition, we will review some central ideas from graph theory. This chapter is mainly based on M. A. Steel (2016).

A *directed graph* or *digraph* is a pair $G = (V, E)$ of sets V and $E \subseteq V \times V$. We call the elements of V *vertices* and the elements of E *directed edges* or just *edges*, if it is not evident which graph we are looking at we might write $V(G)$ and $E(G)$. An edge (u, v) for $u, v \in V$ is said to start at u and end at v. Digraphs are inherently visual structures and can be depicted by drawing a dot for each vertex and an arrow for each edge, directed from the start vertex to the end vertex. If there's a directed edge between two vertices (either direction) they are said to be *adjacent* and the vertices are both *incident* with the edge. If the edge is directed as (u, v) we say that v is a *child* of u, and likewise that u is a *parent* of v. Throughout we will only consider *simple graphs*, meaning that there are no edges from a vertex to itself ("loops") or multiple edges between vertices. Unless otherwise stated, we will assume that the graphs are *finite*, that is $|V| < \infty$. A graph G' is said to be a *subgraph* of G if $V(G') \subset V(G)$ and $E(G') \subset E(G)$. For a subset $V' \subset V(G)$, the *induced subgraph* G' of G is the subgraph with $V(G') = V'$ and $E(G') = \{(u, v) \in E(G) \mid u, v \in V'\}$.

For each vertex in a digraph, we can define the *in-degree* and *out-degree* as the number of directed edges ending and starting, respectively, at the vertex. A *path* is a distinct sequence of vertices $v_1, v_2, ..., v_k$ such that v_i and v_{i+1} are adjacent for $i < k$. It is a *directed path* if $(v_i, v_{i+1}) \in E$ for $i < k$. A *cycle* of length k is a path $v_1, ..., v_k$ such that v_1 and v_k are also adjacent. If the path in a cycle is directed and $(v_k, v_1) \in E$ it is a *directed cycle*. A digraph without any directed cycles is called a *directed acyclic graph* or simply a *DAG*.

A classic result of DAGs is that they induce a *partial ordering* on the vertex set. If there is a directed path from $u \in V$ to $v \in V$ we write $u \prec v$. If $u \prec v$ we say that u is an *ancestor* of v and v is a *descendant* of u. Note that not every vertex is comparable under $\prec$ (hence it is only a *partial* ordering), and neither is the ordering necessarily unique. Moreover, for a DAG we know that it contains at least one vertex of out-degree zero and one of in-degree zero, hence these are in some sense maximal respectively minimal elements under $\prec$.

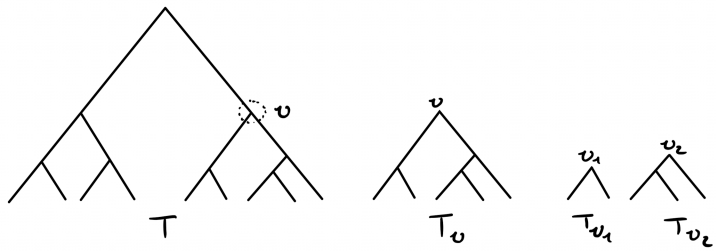

Figure 1: A binary rooted tree T, a re-rooted subtree T_v at $v \in V(T)$ and the two descending subtrees T_{v_1} and T_{v_2} of v.

Trees

Graphical trees exist in several forms depending on the use case. We will consider *binary rooted trees* and some closely related structures, which we will see have some particularly nice biological interpretations. A digraph $T = (V, E)$ is a *rooted tree* if it contains no cycles (directed or not) and for a chosen vertex $\rho \in V$ called the *root,* all other vertices are descendants of the root. *Leaves* are the vertices of out-degree zero, the set of leaves are called $\breve{V}$, and all other vertices are called *interior* and are denoted by $\mathring{V}$. Edges are called *pendant* if they are incident with a leaf while the remaining edges are called *interior*. A rooted tree in which each non-leaf vertex has out-degree two is called a *binary rooted tree*.

One particularly relevant feature of rooted trees for modeling the evolution of species is that for any vertex subset $A \subseteq V$, there is a unique vertex $v \in V$ such that $v \prec a$ or $v = a$ for all $a \in A$, which is maximal under $\prec$, i.e. if there is another vertex w such that $w \prec a$ or $w = a$ for all $a \in A$ then $w \prec v$. We call this vertex v as the *most recent common ancestor of A in T* and write $v = \mathrm{MRCA}_T(A)$.

For a given internal vertex $v \in \mathring{V}(T)$, we let the *re-rooted subtree* T_v of v be the induced subgraph by the set $\{u \in V(T) \mid u \succ v \text{ or } u = v\}$ of descendants of v, we will consider v as the root of this subtree. Likewise, we can define the two (not two if the tree is not binary) *descendant subtrees* T_{v_1}, T_{v_2} being the re-rooted subtrees of the children of v. The two descendant subtrees are the induced subgraphs by the vertex sets $\{u \in V(T) \mid u \leq v_1 \text{ or } u = v_1\}$ and $\{u \in V(T) \mid u \leq v_2 \text{ or } u = v_2\}$ where v_1, v_2 are the two children of v. See fig. 1 for an example of both types of subtrees.

2.1 Phylogenetic trees

Large parts of phylogenetics are concerned with inferring the evolutionary history of a specific set of extant and or extinct species, that is questions of the form *"Are humans more closely related to chimpanzees or to cats"*, *"When did animals and plants diverge"* and *"What are the closest extant*

relative to pterodactyls". Such questions can be answered if we knew the phylogenetic tree, representing the evolutionary history of the species.

While the term phylogenetic tree might seem rather intuitive, the literature uses the term in the broadest possible sense about quite different objects. This can lead to much confusion and to wrong conclusions when deriving mathematical results about a particular type of tree. Thus we introduce a well-defined framework of trees which turns out to be particularly relevant for answering questions like those above.

A *rooted phylogenetic X-tree* is a rooted tree $T = (V, E)$ such that $X \subseteq V$ is the set of leaves in T and every interior vertex has out-degree at least two. If T is a binary rooted tree, we have a *binary rooted phylogenetic X-tree* and these are the objects which we will primarily consider. Note that phylogenetic X-tress inherently have labeled leaves by the set X, while we consider all interior vertices as non-labeled. If we are to ignore the labels X we have an unlabeled rooted tree, we will call such a tree a *tree shape* or a *tree topology*.

X can be interpreted as a set of taxa, but phylogenetic methods are also used in other fields, for example in linguistics where one could consider X a set of languages whose relations we wish to analyze. If nothing else is stated and the set X is clear from the context, we will simply refer to a binary rooted phylogenetic X-trees as a *phylogenetic tree* or even simpler, a *phylogeny*.

Equivalent phylogenies and drawings

Two phylogenies T and T' are *equivalent* if there exists a isomorphism $\phi\colon V(T) \to V(T')$ preserving adjacency and leaf labels, i.e. $(u, v) \in E(T) \iff (\phi(u), \phi(v)) \in E(T')$ for all $u, v \in V(T)$ and $\phi(v) = v$ for $v \in X$, and we write $T \cong T'$. We will not distinguish between equivalent phylogenies, so formally, when working with a phylogeny T, we are referring to the equivalence class of phylogenies under the isomorphism ϕ of which T is a representative. The set of all binary rooted phylogenetic X-trees, up to equivalence, is denoted $\mathrm{RB}(X)$. If the leaves are labeled $X = \{1, 2, ..., n\} =: [n]$ we will just write $\mathrm{RB}(n)$, and the number of phylogenies with n leaves is then denoted $\mathrm{rb}(n)$.

Note that while we will often draw a phylogeny (and trees in more generality) in the plane, this is not a unique representation. The drawing is formally called a *planar embedding* of the tree if no edges cross. In contrast to general graphs such planar embeddings always exist for rooted trees, but they may not be unique (see fig. 2). Hence this induces another equivalence relation, namely that two tree drawings are equivalent if they are planar embeddings of the same tree. Hence we consider a tree drawing a representative of the equivalence class of equivalent planar embeddings and thus of the tree itself.

Equivalent trees and equivalent planar embeddings tell the same story of the evolution of the extant species in X, i.e. it does not matter for the

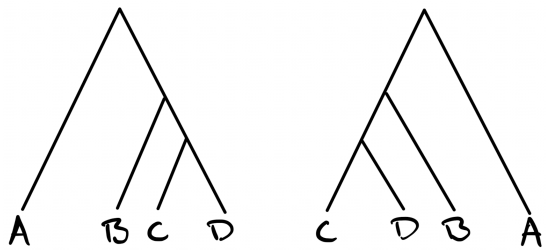

Figure 2: Equivalent planar embeddings of the phylogeny defined in eq. (2)

interpretation of the phylogeny what labels are put on the past lineages or whether species A are to the left or right of species B in the drawing.

Combinatorics and topology of phylogenies

In addition to being graphical structures, trees are also intrinsically combinatorial structures, and a large part of the literature on trees concerns the number of different types of trees (Felsenstein 1978). Some initial counting results on phylogenies with $n \geq 2$ leaves is that they have $2n-2$ edges, $n-1$ interior vertices and that the total number of phylogenies is

$$\mathrm{rb}(n) = (2n-3) \cdot (2n-5) \cdot \cdots \cdot 3 \cdot 1 = (2n-3)!! = \frac{(2n-2)!}{(n-1)!2^{n-1}} \tag{1}$$

Even for a moderately sized X this results in an overwhelming number of trees as illustrated in table 1.

Table 1: Number of phylogenies with n leaves for $n = 2, 5, 10, 20, 40$

n	$\mathrm{rb}(n) = (2n-3)!!$
2	1
5	105
10	34459425
20	8200794532637891559375
40	1009847364737869270905302433221592504062302663202724609375

As well as counting the number of trees, we could also count particular topological features on a given tree, and the number of trees having a given feature. We define a *symmetry vertex* to be an interior vertex such that the two descendant subtrees of the vertex have the same tree shape. Symmetry vertices in which the descendant subtrees are singleton leaves are called *cherries*. For a given phylogeny (or equivalently tree shape) T, we denote the number of symmetry vertices by $s(T)$ and the number of cherries by $c(T)$.

The trees with the lowest number of cherries $c(T) = 1$ are called *caterpillar trees*. The maximal number of cherries is naturally $\lfloor n/2 \rfloor$, and a particular type of tree reaching this bound is *perfect trees*, which only exists

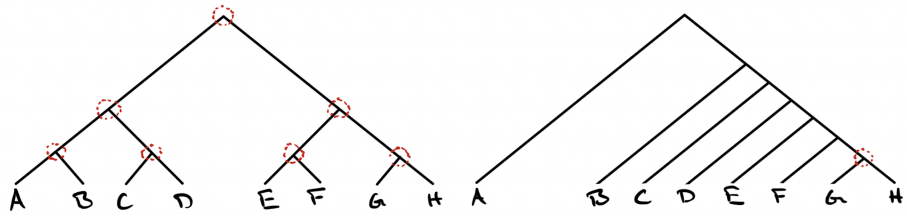

Figure 3: A perfect phylogeny (left) and a caterpillar phylogeny (right) both on $X = \{A, B, ..., H\}$, all symmetry vertices are marked with red.

if n is even, in which every interior vertex is a symmetry vertex (see fig. 3). Symmetry vertices determine which phylogenies we can distinguish from each other when disregarding the leaf labels, we get that the number of phylogenies with n leaves which have the same tree shape τ is given by $n!/2^{s(\tau)}$. And of course if $\mathrm{TS}(n)$ is the space of tree shapes with n leaves, we have that $\sum_{\tau \in \mathcal{T}_n} n!/2^{s(\tau)} = (2n-3)!!$.

In contrast to $\mathrm{rb}(n)$, there is no closed-form equation for the number of tree shapes with a given number of leaves, instead one has to use recursive or asymptotic results. The number of tree shapes with $n = 10$ leaves is 98. So it is evident that the vastness of $\mathrm{rb}(n)$ stems from the leaf labeling and the distinction this induces between trees (Cavalli-Sforza and A. W. F. Edwards 1967).

The Newick format

The usual representation of a phylogeny as the collection of its vertices and edges is not very convenient in practice. Rather one would often use the Newick format[1] representing hierarchical coalescence between elements of X. If $X = \{A, B, C, D\}$ one possible phylogeny is $T = (V, E)$ given by the vertex and edge set, where we have to arbitrarily label the internal vertices as well,

$$\begin{aligned} V &= \{v_1, v_2, v_3, A, B, C, D\}, \\ E &= \{(v_1, A), (v_1, v_2), (v_2, B), (v_2, v_3), (v_3, C), (v_3, D)\}. \end{aligned} \tag{2}$$

With the Newick format, we would just state which leaf nodes are coalescing in a hierarchical parenthesis system, $(A, (B, (C, D)))$. This format is both convenient as it is very compact in comparison to eq. (2), but it also escapes any labeling of the internal nodes which are irrelevant anyways under the equivalence relation defining $\mathrm{RB}(X)$. Actually, the Newick format of a tree is not unique as it represents a planar embedding of the phylogeny, not the phylogeny itself. Thus both $(A, (B, (C, D)))$ and $(((C, D), B), A)$ represent T, but correspond to two different planar embeddings (see fig. 2).

[1] There has been no formal publication of the Newick format, but Joe Felsenstein has written a short presentation at `https://evolution.genetics.washington.edu/phylip/newicktree.html`.

Reconstructed and complete trees

Suppose, as is often the case, that X is a set of species that we have sampled from the currently extant population of species, and we have inferred a rooted binary phylogenetic X-tree from this set. This phylogeny is called *reconstructed* as it does only contain the lineages that are extant and sampled at present as well as those inferred to be ancestral to the sampled extant species (Nee, May, and Harvey 1994). On the other hand, if X contains the sampled extant species as well as all related extinct species, the inferred phylogeny is called *complete*. Actually inferring a complete phylogeny is seldom possible as we do not have data from the past[2], so our methods will always assume that we have a sample of extant species, and will create probability measures on spaces of reconstructed trees.

2.2 Other trees

Phylogenetic X-trees are often biologically relevant, they depict evolution as happening through time due to the direction of edges, ending up with the set of extant species which we have sampled. However, they might not always be the easiest tool to work with, and sometimes simpler or more complicated structures are more appropriate to answer particular questions. Thus we introduce a few related tree structures which will turn out to be needed in the sequel.

Oriented trees

In chapter 4 we will see that many models generating phylogenetic trees actually produce an *oriented binary tree*. An oriented binary tree is a rooted binary tree in which the child of each node is assigned an arbitrarily ordered label, in phylogenetics it is customary to label them *left* and *right*, but we will assign them with either 1 or 2, hence we can talk about the "first" and "second" child species, even though there might not be a temporal ordering, see fig. 4 for an example of an oriented tree. Oriented trees are also called plane trees, as they correspond to a specific planar embedding. We call the space of all oriented binary trees OR and by $\mathrm{OR}(n)$ we mean the oriented binary trees with n leaves, (Stanley 1999, Example 5.3.12) shows there are C_{n-1} oriented trees with n leaves, where C_{n-1} is the $n-1$'st Catalan number, hence we can write,

$$|\,\mathrm{OR}(n)| = C_{n-1} = \frac{1}{n-1+1}\binom{2(n-1)}{n-1} = \frac{(2n-2)!}{n!(n-1)!}.$$

Note that the leaves of an oriented tree are not labeled in the same arbitrary way as phylogenies are, here the leaf labels completely determine the tree, while there is no information in the labels of a phylogeny. Hence if we want to construct a phylogeny from an oriented tree, we need to

[2] Inferring complete tree can be possible if we include data from fossils (M. J. Donoghue et al. 1989; Stadler 2010)

ignore the orientations, thus obtaining a tree shape, and assign this tree shape arbitrary leaf labels. So if we take a tree shape τ and consider an internal vertex $v \in \mathring{V}(\tau)$ of it, there are two ways to orient the children of v such that the resulting oriented descendant subtrees have different tree shapes, unless, of course, if v is a symmetry vertex in which case the two orientations would result in the same tree shape. Thus if $s(\tau)$ is the number of symmetry vertices in a tree shape τ with n leaves, there are $2^{n-1-s(\tau)}$ oriented trees with that tree shape. As described in section 2.1, for a tree shape τ with n leaves there are $n!/2^{s(\tau)}$ phylogenies with that tree shape, so for a given oriented tree there are

$$\frac{n!}{2^{s(\tau)}2^{n-1-s(\tau)}} = \frac{n!}{2^{n-1}}$$

possible phylogenies which have the same tree shape. We also see that $\mathrm{rb}(n) = n!/2^{n-1}|\,\mathrm{OR}(n)|$.

We are not particularly interested in oriented trees as such, but they provide a nice structure to work with mathematically, and the result above shows that if we know the probability of an oriented tree with a given tree shape, we can find the probability of a phylogeny with that tree shape by multiplying by $2^{n-1}/n!$ (Ford, Matsen, and Stadler 2009; Stadler 2010).

Planted trees

Planted trees are mostly an aesthetic extension of a rooted tree, which will be more natural when working with evolutionary processes through time in chapter 4. For a rooted tree T the associated planted tree has an extra edge going into the root coming from a *phantom vertex* which we do not formally include in the vertex set. This extension gives us the nice result that every vertex has in-degree 1 and we thus have a one-to-one correspondence between edges and nodes, but more importantly the root edge can also be assigned a length, for example when considering the planted tree a time tree. The root edge can only be placed at one place in the tree, so if not considering branch lengths there is a bijection between a rooted tree and its corresponding planted tree. In particular, every counting result will carry over to the planted trees.

Time trees

As already noted, phylogenies show evolution as happening through time, but the time unit is not defined, and there is no temporal order of the speciation events. To answer temporal questions we thus need to introduce *time trees*. Let τ be a planted binary rooted tree. For each edge $e \in E(\tau)$ (including that between the phantom vertex and the root) define the associated branch length $t_e \in \mathbb{R}_+$ which we will interpret as the time during which a given lineage exists. The tuple $T = (\tau, \{t_e\}_{e \in E(\tau)})$ is then called a *time tree*.

We could equivalently associate each edge with a *branching time* $\delta_v \in \mathbb{R}_+$ indexed by the vertex the edge points towards. We assume that $\delta_v < \delta_u$

if $v \prec u$ and let the branch length of an edge $(v,u) \in E(\tau)$ be $\delta_u - \delta_v$ and the branch length of the edge going into the root be δ_ρ, that is we consider the branching time of the phantom vertex to be 0. With the collection of branching times, we get a time tree $T = (\tau, \{\delta_v\}_{v \in V(\tau)})$. We write δ_T for the set of branching times of the time tree. When considering descendant subtrees T_{v_1}, T_{v_2} of a vertex v in a time tree T, we look at them as time trees with origin at the branching time δ_v, i.e we could now consider v the phantom node of each of the subtrees. Thus, note that for a descendant subtree T_{v_1}, the set of branching times $\delta_{T_{v_1}}$ also contains the branching time of v.

Ulam-Harris trees

Consider the set

$$\mathscr{I}_2 = \bigcup_{n=0}^{\infty} \{1,2\}^n$$

with $\{1,2\}^0 = \{0\}$, which we will call the *binary Ulam-Harris family space*. That is, elements of $\mathscr{I}$ are of the form $\iota = \langle i_1 i_2 ... i_k \rangle$ where $i_j \in \{1,2\}$ for each $j \leq k$ or $\iota = \langle 0 \rangle$. For two elements $\iota_1, \iota_2 \in \mathscr{I}_2 \setminus \{\langle 0 \rangle\}$ we write $\iota_1 \iota_2$ for the concatenated element having first ι_1's and then ι_2's coordinates, and when concatenating with $\langle 0 \rangle$ we have $\langle 0 \rangle \iota = \iota = \iota \langle 0 \rangle$ for any $\iota \in \mathscr{I}_2$. We can further define some operators on the space, first m and r defined such that for any $\iota = \langle i_1 i_2 ... i_k \rangle$ we have

$$m\iota = \langle i_1 i_2 ... i_{k-1} \rangle, \quad r\iota = \langle i_k \rangle$$

and if $j < k$, we define a_j and d_j by

$$a_j \iota = \langle i_1 i_2 ... i_j \rangle, \quad d_j \iota = \langle i_{j+1} ... i_k \rangle.$$

Hence we in particular have $m\iota\, r\iota = \iota$, $a_j \iota\, d_j \iota = \iota$ and $m^k \iota = \langle 0 \rangle$.

We will consider this set as the vertex set of an infinite (but countable) rooted tree $\mathscr{U}$ with edge set $E(\mathscr{U}) = \{(\iota_1, \iota_2) \mid \iota_1 = m\iota_2\}$. The vertices of this tree then completely encode their own ancestry without direct specification of the edges; the i_k'th child of the i_{k-1}'th child of the ... of the i_2'th child of the i_1'th child of $\langle 0 \rangle$ is denoted $\langle i_1 i_2 ... i_k \rangle$. The tree is clearly binary as each ι has exactly the children $\iota\langle 1 \rangle$ and $\iota\langle 2 \rangle$, and this naming of the children makes it evident that it is also an oriented tree. We will furthermore consider $\mathscr{U}$ a planted tree, such that there is an edge going into $\langle 0 \rangle$.

As this construction might suggest, the set of oriented binary trees OR can be identified with the finite rooted subtrees T of $\mathscr{U}$ that satisfy

$$\iota\langle i \rangle \in V(T) \implies \iota\langle j \rangle \in V(T) \quad \text{for } j \leq i,$$

if we further require that $\breve{V}(T) = n$, we obtain the set $\mathrm{OR}(n)$. Thus, as we could encode phylogenetic X-trees using the Newick format, we can encode oriented binary trees by (appropriate) subsets of the binary Ulam-Harris family space, oriented trees with the Ulam-Harris labeling are called *Ulam-Harris trees*. See figs. 4 and 6 for examples of Ulam-Harris trees.

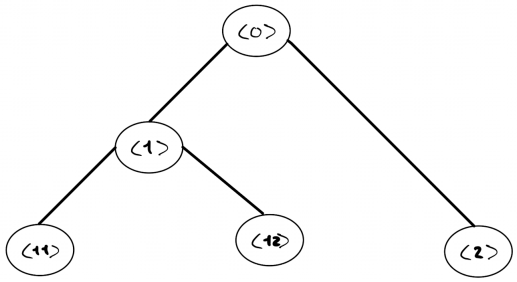

Figure 4: An oriented tree with the Ulam-Harris encoding.

It will turn out to be helpful in the sequel to consider re-rooted subtrees of $T \in \mathrm{OR}$, in which we consider the subtree descending from a new root $\kappa = \langle i_1 i_2 ... i_j \rangle$ that is, $T_\kappa = \{d_j \iota \mid \iota \in T,\, a_j \iota = \kappa\}$.

3 Probability on discrete tree spaces

Our main objective is to define and use probability measures on tree spaces, so let us introduce some probability notation. A random tree is a measurable mapping $\mathscr{T} : \Omega \to \mathbb{T}$ where $(\Omega, \mathcal{F}_\Omega, \mathbb{P})$ is some background probability space and $(\mathbb{T}, \mathcal{F}_\mathbb{T})$ is a measurable tree space. We then have an induced probability measure on $(\mathbb{T}, \mathcal{F}_\mathbb{T})$ given by $\mathbb{P} \circ \mathscr{T}^{-1}$ also called the distribution or model of $\mathscr{T}$. Note that whenever the tree space is finite or countable, we can (often silently) choose the σ-algebra to be the power set $2^\mathbb{T}$ and the distribution is given completely by the probability of singleton trees.

3.1 The uniform model

We will usually use $\mathrm{RB}(n)$ or some extension of it as our tree space, and with the knowledge that $\mathrm{rb}(n) = (2n-3)!!$ we have a trivial example of a distribution on $\mathrm{RB}(n)$,

Definition 3.1 (The uniform model)**.** We say the random tree $\mathscr{T}$ is generated under the uniform model P_U (also called the "proportional-to-distinguishable-arrangements model") on $\mathrm{RB}(n)$ if

$$\mathbb{P}(\mathscr{T} = T) = P_U(T) = \frac{1}{(2n-3)!!}, \quad \text{for } T \in \mathrm{RB}(n).$$

Simply picking a phylogeny at random does not directly draw on any knowledge of the actual evolutionary process which the tree represents. There have been suggested biological motivations for the uniform model (Blum and François 2006), but none of which are as natural and simple as the model in the next section. Furthermore, comparisons with empirical phylogenies show that the uniform model generally produces too unbalanced trees, see section 3.4.

3.2 The Yule-Harding model

Instead, we could try to come up with a distribution over $\mathrm{RB}(n)$ which is directly deduced from a description of the evolutionary process. Accordingly, consider a single species that split into two, such that we initially have the tree shape on two leaves. Now to mimic the evolutionary process we choose one of the two leaves at random and let it split, such that we now have three leaves. This process is repeated until we have a tree shape with n leaves, we then randomly assign labels to the leaves from $[n]$. The resulting tree is a random sample from the *Yule-Harding (YH) model or distribution*,

Proposition 3.2. *If the random tree $\mathscr{T}$ is generated under the YH model P_{YH}, we have*

$$\mathbb{P}(\mathscr{T} = T) = P_{\mathrm{YH}}(T) = \frac{2^{n-1}}{n! \prod_{v \in \check{V}(T)} \ell_v - 1}, \quad \textit{for } T \in \mathrm{RB}(n).$$

where $\ell_v = \left|\{u \mid u \in \check{V}(T), v \prec u\}\right|$ is the number of descendant leaves of v.

The Yule-Harding model has an equivalent interpretation under the discrete coalescent process[3] where we start with n lineages and merge ("*coalesce*") any of the $n(n-1)/2$ pairs of lineages uniformly at random, resulting in $n-1$ lineages, this continues until we only have one lineage remaining. The resulting tree is a random tree with the Yule-Harding distribution. Under other more complicated evolutionary models, including both speciation and extinction, it turns out that the distribution of their resulting reconstructed phylogenies is exactly the YH distribution, see section 4.5 and D. Aldous (1996).

3.3 The Aldous' β-splitting model

Another model was "pulled out of thin air" by D. Aldous (1996) with the sole purpose of generalizing the Yule-Harding and uniform models from a purely mathematical point of view. Consider a symmetric distribution s_n on $[n-1]$, i.e. $s_n(j) = s_n(n-j)$ for all $j \in [n-1]$, we call these *splitting distributions*. Now at the root, we partition the n species into two sets, which we consider as the leaves of the two descendant subtrees, one of size $J \sim s_n$ and one of size $n-J$. The elements in each set are chosen uniformly at random from the $\binom{n}{J}$ possibilities. This construction is then applied recursively to each resulting set until each partition is of size 1. This results in a random phylogeny and we call this general construction the *Aldous' splitting model*.

The choice of s_n gives rise to different models, one particularly versatile is the following. Consider the (one-parameter) beta distribution $\mathrm{B}(\beta)$ for

[3] Kingman's coalescent Kingman (1982) is a continuous time process much used in population genetics.

$\beta \in (-1, \infty)$ with density,

$$f(y) = \frac{\Gamma(2\beta+2)}{\Gamma(\beta+1)^2} y^\beta (1-y)^\beta, \quad \text{for } y \in (0,1)$$

and let $Y \sim \mathrm{B}(\beta)$. Now place the species in X uniformly at random on the unit interval and sample a point y from Y, this will partition X into a set of size J and $n-J$ where J conditional on $Y = y$ follows a binomial distribution $\mathrm{Binom}(n, y)$, and hence marginal distribution

$$\mathbb{P}(J = j) = \int_0^1 \binom{n}{j} y^j (1-y)^{n-j} f(y) \, \mathrm{d}y$$

As we always want the two resulting sets to be non-empty, we condition on $J \neq 0, n$, and arrive at a candidate splitting distribution,

$$\begin{aligned} s_n(j) = \mathbb{P}(J = j \mid J \neq 0, n) &= \frac{\mathbb{P}(J = j)}{\mathbb{P}(J \neq 0, n)} \\ &= a_n(\beta)^{-1} \frac{\Gamma(\beta+j+1)\Gamma(\beta+n-j+1)}{\Gamma(j+1)\Gamma(n-j+1)}, \quad \text{for } j \in [n-1]. \end{aligned} \tag{3}$$

where $a_n(\beta)$ is a normalizing constant, making s_n a probability mass function. Although the β-distribution requires $\beta \in (-1, \infty)$, eq. (3) is well-defined for $\beta \in (-2, \infty)$ and converges for $\beta \to -2^+$, so we will consider s_n a splitting distribution for $\beta \in [-2, \infty)$. The model on random phylogenies given by this splitting distribution is the *Aldous' β-splitting model*, which induces a distribution on $\mathrm{RB}(n)$ as follows

Theorem 3.3. *If the random tree $\mathscr{T}$ is generated under the β-splitting model, the induced distribution P_β of $\mathscr{T}$ satisfies the recursion with the descendant subtrees of its root,*

$$\mathbb{P}(\mathscr{T} = T) = P_\beta(T) = 2 s_n(j) \binom{n}{j}^{-1} P_\beta(T_1) P_\beta(T_2)$$

for $T \in \mathrm{RB}(n), T_1 \in \mathrm{RB}(j), T_2 \in \mathrm{RB}(n-j)$ where j is the number of leaves in one of the descendant subtrees.

Proof. Let T_1 and T_2 be the two descendant subtrees of T and let them have j respectively $n-j$ vertices, where we may assume that $2j \leq n$, such that T_2 is the largest subtree. Under the β-splitting model the probability of such a split is $2s_n(j)$ if $2j < n$ and s_n if $2j = n$. And under this partition, we can choose the leaf set of size j (and thus also that of size $n-j$) in $\binom{n}{j}$ ways if $2j < n$ and in $\frac{1}{2}\binom{n}{j}$ ways if $2j = n$, taking into account that we can interchange the two descendant subtrees in T if they have the same size. Now the two subtrees follow the β-splitting model independently leading to the stated recursion. □

As was the target, the β-splitting model has the Yule-Harding model, the uniform model, and several other well-known models as special cases. If we let $\beta = 0$ we obtain the Yule-Harding model

$$s_n(j) = a_n(0)^{-1} = \frac{1}{n-1}$$

assigning a uniform distribution to the size of descendant subtrees. For $\beta = -3/2$ we obtain the uniform model, which can be shown to have splitting distribution

$$s_n(j) = \frac{1}{2}\binom{n}{j}\frac{\mathrm{rb}(j)\,\mathrm{rb}(n-j)}{\mathrm{rb}(n)}$$

Using induction one can easily check that the recursion in theorem 3.3 is satisfied by these two models with their respective choice of β, but the true versatility of the β-splitting model is that it induces a whole continuum of splitting models, some of which might fit empirical data better than the somewhat heuristically derived Yule process and the uninformative uniform model. It turns out that in a certain sense β controls the balance of the produced trees.

3.4 Tree balance

There exist many measures of balance in a phylogeny, but they all try to capture how evenly leaves descending from each interior vertex are split between the descendant subtrees. We have already seen the two extremes of balance, namely the perfect tree, which is perfectly balanced, and the caterpillar tree, which is as unbalanced as it gets. But we want more generally to measure the expected balance of trees produced by a stochastic model. A natural starting point is exactly the splitting distribution from before as these describe the distribution of the number of leaves on descendant subtrees. If the splitting distribution gives a high probability for splits like $n-1:1$ (meaning that $J = n-1$ and $n-J = 1$) it will produce very unbalanced trees, while if it gives high probability to splits like $n/2 : n/2$ (assuming that n is even) we get very balanced trees. Using the β-splitting model we see that low values of β give us unbalanced trees, while higher values result in more balanced trees (see fig. 5).

Blum and François (2006) and D. J. Aldous (2001) has considered the fit of β-splitting models to empirical trees, from TreeBASE (Piel, M. Donoghue, and Sanderson 2009) and other sources, and have found that the empirical trees seem to more unbalanced than those generated by the Yule-Harding model, but more balanced than those generated by the uniform model. They both find that $\beta \approx -1$ is a good fit. While there are some problems with the estimation method in Blum and François (2006) as described in Jones (2011b), the same overall conclusion that empirical trees have a balance somewhere between that of the uniform and the Yule-Harding model is reached. Moreover the $\beta = -1$ case has some particularly nice mathematical properties (D. Aldous 1996).

4 Tree evolution

As has been subtly noted several times, evolution is inherently a process happening through time, so discrete phylogenies as above will never tell

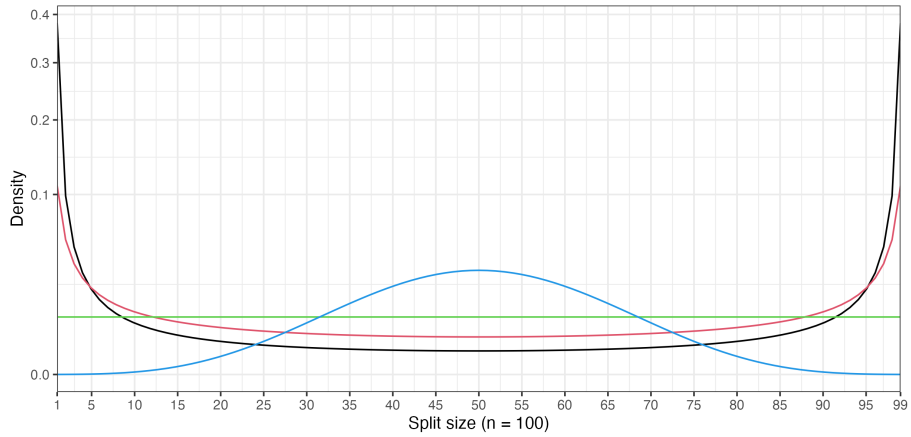

Figure 5: β-splitting distributions with $n = 100$ for four different β values. $\beta = -3/2$ (uniform) in black, $\beta = -1$ in red, $\beta = 0$ (Yule-Harding) in green and $\beta = 10$ in blue.

the full story, in particular, if we want to answer questions with a temporal component.

Since the initial paper by Yule (1925) many stochastic processes have been suggested to model the macroevolutionary process. The Yule process is a continuous time version of the Yule-Harding model of section 3.2 which only contains speciations. But speciations is not the only event defining the evolution of species, extinctions are often as important, hence the birth-death model, where species can both speciate and go extinct has been the standard model for the evolution of species. We will present a generalization of both the Yule process and the birth-death model, which is rooted in the branching process literature.

4.1 A branching process point of view

Branching processes consider systems of objects which in time can recursively multiply or disappear under a set of stochastic laws, the defining property being that once an object has split, each descendant object behaves independently of its ancestral object and any other objects in the system. This seems like a natural framework to model macro-evolution, considering species as the objects which might speciate (split/multiply) or go extinct (disappear). We will present an age-dependent branching process, named the binary Bellman-Harris process (Bellman and T. Harris 1952), which we will see is a generalization of several standard macro-evolutionary models. A nice byproduct of the branching process formulation is that we get a direct formal description of the tree induced by the process.

Informally the process initiates with a single species at time $t = 0$, this species has a lifetime L with distribution function $G(t)$. At the end of the

species' lifetime, which we will call the *termination* of the species, it either speciates (produces two new species) with probability q or goes extinct (produces no new species) with probability $1-q$. The eventual new species exist and terminate according to the same distribution as their ancestor, and they do so independently from their ancestor and any other species. The same procedure continues until there are no extant species.

Family history

Formalizing this, we use ideas from (T. E. Harris 1963; Jagers 1989; Jagers and Sagitov 2008) defining an explicit background probability space $(\Omega, \mathscr{A}, \mathbb{P})$ from which we can define both the branching process itself (counting the number of extant species at time t) and what is called the *family tree*, which will come to represent a phylogeny. Consider again the binary Ulam-Harris family space of section 2.2

$$\mathscr{I}_2 = \bigcup_{n=0}^{\infty} \{1,2\}^n$$

with $\{1,2\}^0 = \{0\}$. We will think of the elements $\iota \in \mathscr{I}_2$ as possible species in the following way, $\iota = \langle i_1 i_2 ... i_k \rangle$ is *the i_k'th daughter species of the i_{k-1}'th daughter species of the ... of the i_2'th daughter species of the i_1'th daughter species of the ancestral root species* $\langle 0 \rangle$. That is, we could have an element $\iota = \langle 122 \rangle$ which is *the second daughter species of the second daughter species of the first daughter species of the ancestor species.* Note again that as both new species are born at the same time there is no natural order between them. This also shows that there is no deeper purpose in labeling species using numbers, in fact, it is common to label them as "left" and "right" in the phylogenetic literature (Ford, Matsen, and Stadler 2009).

For the ancestor $\langle 0 \rangle \in \mathscr{I}_2$ we construct a probability space $(\Omega_0, \mathscr{A}_0, \mathbb{P}_0)$. Ω_0 should contain all possible "life trajectories" of the ancestor, in the sense that all necessary properties of the ancestor are measurable functions on $(\Omega_0, \mathscr{A}_0)$. In our context, the life of the ancestor is given by its life length and whether it speciates or goes extinct at the end of its life, this could be encoded as a non-negative real number ℓ and a number $\nu \in \{0,2\}$, i.e. $\Omega_0 = \mathbb{R}_+ \times \{0,2\}$. We can equip this space with the product σ-algebra $\mathscr{A}_0 = \mathcal{B}_+ \otimes 2^{\{0,2\}}$, where $\mathcal{B}_+$ is the Borel σ-algebra on $\mathbb{R}_+$, and let the probability measure $\mathbb{P}_0$ be defined by

$$\begin{aligned} &\mathbb{P}_0(L \leq t) = G(t), \quad \text{for } t \geq 0, \\ &\mathbb{P}_0(N = 2) = 1 - \mathbb{P}_0(N = 0) = q \end{aligned}$$

where $L : \Omega_0 \to \mathbb{R}_+$ and $N : \Omega_0 \to \{0,2\}$ are independent random variables representing the lifetime and number of offspring species of the ancestor, i.e. they are projections of Ω_0 onto $\mathbb{R}_+$ and $\{0,2\}$ respectively. We assume that $G(0) = 0$ ensuring that the process defined in section 4.1 is almost surely finite and that $G(t)$ is differentiable, i.e. the lifetime distribution has

a density $g(t)$. If $p > 1/2$ we call the process supercritical, if $p < 1/2$ we call the process subcritical, and if $p = 1/2$ the process is called critical.

As we assume that each species behaves i.i.d. like their ancestor, we can construct the full probability space $(\Omega, \mathscr{A}, \mathbb{P})$ as the countable product space

$$(\Omega, \mathscr{A}, \mathbb{P}) := \left(\prod_{\iota \in \mathscr{I}_2} \Omega_\iota, \bigotimes_{\iota \in \mathscr{I}_2} \mathscr{A}_\iota, \bigotimes_{\iota \in \mathscr{I}_2} \mathbb{P}_\iota \right)$$

where $(\Omega_\iota, \mathscr{A}_\iota, \mathbb{P}_\iota)$ are copies of the ancestor's probability space. As for the ancestor we get random variables L_ι, N_ι being i.i.d. copies of L and N. Note that we generally disregard the ι index whenever $\iota = \langle 0 \rangle$ as is customary in the branching literature.

An element $\omega \in \Omega$ now describes the whole evolutionary development stemming from the ancestor but also contains unnecessary additional information such as the number of offspring species and the lifetime of a species which are never realized, for example, if $N_{\langle 1 \rangle}(\omega) = 0$ then all species of the form $\langle 1\, i_2 ... i_k \rangle$ are never actually born, hence we do not care about any of their properties. Keeping this unnecessary information in Ω however, allows the simple product space interpretation.

The k'th generation $I_k(\omega)$ is the set of species that are actually realized for a given ω and has k ancestors. We can define these generations recursively,

$$\begin{aligned} &I_0(\omega) = \{\langle 0 \rangle\}, \quad I_1(\omega) = \{\langle i \rangle \in \mathscr{I}_2 \mid i \leq N(\omega)\} \\ &I_k(\omega) = \{\langle i_1 ... i_k \rangle \in \mathscr{I}_2 \mid \langle i_1 ... i_{k-1} \rangle \in I_{k-1}(\omega),\, i_{k-1} \leq N_{\langle i_1 ... i_{k-1} \rangle}(\omega)\} \end{aligned}$$

for $k \geq 2$. The set of all realized objects $I(\omega) = \bigcup_{k=0}^{\infty} I_k(\omega)$ is called the *family*, which is of course a, possibly infinite, subset of the binary Ulam-Harris space $I(\omega) \subseteq \mathscr{I}_2$, i.e. an oriented tree.

The branching process

What is usually called *the branching process* is the process counting the number of extant species over time. This can now be defined as a (measurable) function on the family history space. We will count an individual ι at time $t \geq 0$ if it is actually realized and if it is born before or at time t and has not died at time t, we define the event on which this happens for all $\iota \in \mathscr{I}_2$ and $t \geq 0$

$$A_\iota(t) = \left\{ \omega \in \Omega \,\middle|\, \iota \in I(\omega),\, \sigma_\iota(\omega) \leq t < \delta_\iota(\omega) \right\}$$

where $\sigma_\iota = L + L_{\langle i_1 \rangle} + \cdots + L_{m\iota}$ for $\iota \neq \langle 0 \rangle$ and $\sigma_{\langle 0 \rangle} = 0$ is the speciation time when ι was born and $\delta_\iota = \sigma_\iota + L_\iota$ is the termination time of $\iota \in I(\omega)$. The indicator function on each of these events forms the basis for counting the number of extant species, $Z_\iota(t, \omega) := \mathbf{1}_{A_\iota(t)}(\omega)$. Hence to get the branching process we just sum all these indicator functions,

$$Z(t, \omega) := \sum_{\iota \in \mathscr{I}_2} Z_\iota(t, \omega)$$

$Z(t,\cdot) : \Omega \to \mathbb{N}_0$ is a random variable for given $t \geq 0$ and will often be denoted $Z(t)$, the collection of all these forms a stochastic process denoted by $Z = \{Z(t)\}_{t\geq 0}$.

The branching tree

In the same way, we can define the possibly infinite random time tree generated by a family history ω as a (measurable) function $\mathscr{T}$ on the family history space, simply by considering the family as an Ulam-Harris tree and the branching times as given by the cumulative lifetimes δ_ι for each ι

$$\mathscr{T}(\omega) := (I(\omega), \{\delta_\iota(\omega)\}_{\iota\in I(\omega)})$$

note that this is an oriented and planted time tree.

$\mathscr{T}$ denotes the complete random tree which may extend indefinitely and contains both extinct and extant lineages. In practice, we are interested in the reconstructed tree at a time $t \geq 0$ since the origin of the process as described in section 2.1. Let us first define the complete tree at some sampling time t, $\mathscr{T}_c(t) := (I_c(t), \{\delta_\iota^t\}_{\iota\in I_c(t)})$ where

$$\begin{aligned} I_c(t,\omega) &= \{\iota \in I(\omega) \mid \sigma_\iota(\omega) \leq t\} \\ \delta_\iota^t(\omega) &= \delta_\iota(\omega) \wedge t, \quad \text{for } \iota \in I_c(t,\omega). \end{aligned}$$

Now, let the reconstructed tree $\mathscr{T}_r(t)$ be the complete tree, where species which have gone extinct before time t, i.e. if $\delta_\iota < t$ and $N_\iota = 0$, are deleted and likewise are the edge going into them. We then get a tree with wrong orientational labels and with species whose out-degree is one, but relabeling the tree and suppressing out-degree one vertices gives us an oriented time tree, where the speciation times are accumulated for the suppressed vertices. See fig. 6 for an example of a complete tree and the process of getting the reconstructed tree. A more detailed exposition on time-tree probability spaces with a discussion of the appropriate σ-algebras and dominating measures can be found in Lambert (2010).

4.2 Some results for the binary Bellman-Harris process

We wish to compute distributions over reconstructed phylogenetic trees generated by the general Bellman-Harris process. The approach is similar to that of (Gernhard, Hartmann, and M. Steel 2008; Jones 2011a) but we concentrate on the situation with only one type and we will consider time as moving forward from the origin of the process. To aid the derivation of the tree distributions, we need some intermediary results.

Let Z and $\mathscr{T}_r(t)$ be the branching process and the reconstructed time-tree as described in section 4.1. Let $p_n(t) = \mathbb{P}(Z(t) = n)$. We will in particular be interested in $p_0(t)$ and $p_1(t)$, so we will find integral equations which these two probabilities satisfy. Let $F(s,t) = \mathbb{E}[s^{Z(t)}]$ be the generating function of Z for $s \in \overline{D}(0;1) = \{z \in \mathbb{C} \mid |z| < 1\}$. T. E. Harris (1963,

Theorem VI 7.1) shows that the generating function satisfies the following integral equation

$$F(s,t) = s\big(1 - G(t)\big) + \int_0^t h\big(F(s,t-u)\big)\,\mathrm{d}G(u) \tag{4}$$

where $h : \overline{D}(0;1) \to \mathbb{C}$ is the generating function of N, hence for our binary Bellman-Harris process it is given by

$$h(s) = \sum_{n=0}^{\infty} \mathbb{P}(N = n)s^n = (1-q)s^0 + qs^2 = (1-q) + qs^2. \tag{5}$$

So the integral equation eq. (4) can be written as

$$F(s,t) = s\big(1 - G(t)\big) + (1-q)\int_0^t \mathrm{d}G(u) + q\int_0^t F(s,t-u)^2\,\mathrm{d}G(u) \tag{6}$$

$$= s\big(1 - G(t)\big) + (1-q)G(t) + q\int_0^t F(s,t-u)^2\,\mathrm{d}G(u). \tag{7}$$

under the assumption that $G(0) = 0$.

Now, generating functions completely determine the point probabilities $p_n(t)$ at any time t through its power-series coefficients

$$p_n(t) = \mathbb{P}(Z(t) = n) = \frac{F^{(n)}(0,t)}{n!} \tag{8}$$

so in particular we can find integral equations for $p_0(t)$ and $p_1(t)$,

Theorem 4.1. *The probabilities $p_0(t)$ and $p_1(t)$ satisfies the following integral equations*

$$p_0(t) = (1-q)G(t) + q\int_0^t p_0(t-u)^2\,\mathrm{d}G(u)$$

$$p_1(t) = \big(1 - G(t)\big) + 2q\int_0^t p_0(t-u)p_1(t-u)\,\mathrm{d}G(u)$$

for $t \geq 0$.

Proof. The result follows directly from eq. (8) with $n = 0$ and $n = 1$, assuming that derivatives and integrals are exchangeable.[4] □

For $n \geq 2$, $p_n(t)$ has the following form,

$$p_n(t) = q\int_0^t \sum_{k=0}^{n} p_k(t-u)p_{n-k}(t-u)\,\mathrm{d}G(u) \tag{9}$$

which is shown heuristically in (Jones 2011a) but should be possible to prove using eq. (8) and Faá di Bruno's generalization of the chain rule (Roman 1980).

[4] (Jones 2011a) gives a somewhat more intuitive derivation of the probabilities directly from the process description.

The reconstructed MRCA

Consider the complete tree $\mathscr{T}_c(t)$ and the reconstructed tree $\mathscr{T}_r(t)$ at some sampling time $t \geq 0$. If $Z(t) > 1$ then the $Z(t)$ extant species have a most recent common ancestor (MRCA) in both $\mathscr{T}_c(t)$ and $\mathscr{T}_r(t)$, the MRCA will be $\langle 0 \rangle$ in both trees, but these do not necessarily have the same temporal placement, i.e. if we let the speciation time of the MRCA in the reconstructed tree be $\overset{\star}{\delta}{}^t$ we might have $\overset{\star}{\delta}{}^t_{\langle 0 \rangle} > \delta^t_{\langle 0 \rangle}$, as whole subtrees of the MRCA in $\mathscr{T}_c(t)$ might go extinct. Let us consider the distribution of $\overset{\star}{\delta}{}^t_{\langle 0 \rangle}$, having density $d^t(s)$ for $s \leq t$. More generally, for a descendant subtree started at time $u \geq 0$ let $d^t(s,u)$ be the density of the speciation time of the reconstructed MRCA in that subtree, note that $d^t(s) = d^t(s,0)$. See fig. 6 for an illustration of the difference between δ_ι and $\overset{\star}{\delta}_\iota$. This density satisfies an integral equation akin to those of p_0 and p_1,

Theorem 4.2. *The density of the speciation time of the reconstructed most recent common ancestor $d(s,u)$ satisfies the integral equation,*

$$d^t(s,u) = qg(s-u) + 2q \int_u^s g(w-u) p_0(t-w) d^t(s,w) \, \mathrm{d}w$$

for $0 < u < s \leq t$.

Proof. We present a somewhat vague, but intuitive proof. Consider a subtree started at time u. The time $s > u$ of speciation of the reconstructed most recent common ancestor can arise in two disjoint ways. First, it might actually be the MRCA in the complete subtree, hence there are no terminations until s (with density $g(s-u)$) at which there is a termination, which must be a speciation (with probability q), observed in the reconstructed tree. Otherwise there has been at least one speciation before s, let the first one be at time $w \in (u,s)$ (with density $g(w-u)$). As this speciation is not observed in the reconstructed tree, one of the descendant subtrees must go extinct before time t (with probability $p_0(t-w)$), which one does not matter, and the other one must have a reconstructed MRCA at time s (with density $d^t(s,w)$), note that there might similarly have been further unobserved speciation between w and s. This explains the terms of the integral equation. □

Remark that if the branching process has no extinctions, that is if $q = 1$, the reconstructed MRCA is the actual MRCA, and we get $d^t(s,u) = g(s-u)$. This is also seen easily from theorem 4.2 as $p_0 \equiv 0$ if there are no extinctions, so the integral will always disappear.

Computing the integral equations

To use these results, the integral equations above should have a computable solution. In the special case where the lifetime distribution is an exponential distribution, the solution attains an analytical form, see section 4.5, but in more general this is not guaranteed. Instead, we must

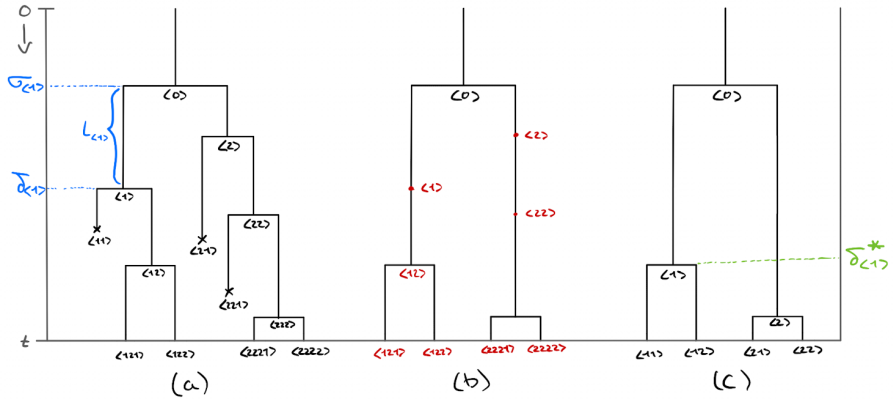

Figure 6: (a) A complete branching tree with extinctions (marked by a cross), σ_ι and δ_ι are illustrated. (b) The complete tree with extinct lineages removed and wrong Ulam-Harris encoding. (c) The reconstructed tree and an illustration of the timing of the reconstructed MRCA δ_ι^*Note that while the most recent common ancestor in the left descendant subtree of $\langle 0 \rangle$ is encoded $\langle 1 \rangle$ its timing is vastly different.

resort to numerical solutions. In Jones (2011a, Theorem 1) it is shown that the equations in theorem 4.1 and eq. (9) has solutions and that the one of theorem 4.2 has a solution if the density g is bounded. For all of the solutions, it holds that they are unique among bounded functions.

An iterative procedure for solving the equations is then proposed in Jones (2011a). Another approach is to use eq. (8) directly and compute the n'th derivatives of the generating function using the Cauchy integral formula as described for a more general branching process in Andersen (2023).

4.3 Tree density

In the following f will denote a generic density function (can also be a density wrt. a counting measure, i.e. a probability mass function), the distribution of which it is a density should be clear from the context. The results of the preceding section give us the ability to compute $p_1(t)$ and $d^t(s,u)$ which in turn gives the distribution of each branch and speciation time in the reconstructed tree. That is, with these quantities we can state the density function of $\mathscr{T}_r(t)$,

Theorem 4.3. *Let $\mathscr{T}_r(t)$ be the reconstructed tree at time $t \geq 0$ of the binary Bellman-Harris branching process. For a realized reconstructed oriented time tree $T^o(t) = (\tau^o(t), \{x_\iota\}_{\iota \in V(\tau^o(t))})$ the density of $\mathscr{T}_r(t)$ is,*

$$f(T^o(t)) = \prod_{\iota \in \check{V}(T^o(t))} d^t(x_\iota, x_{m\iota}) \prod_{\iota \in \check{V}(T^o(t))} p_1(t - x_{m\iota}).$$

Proof. The density of $\mathscr{T}_r(t)$ (and any subtree thereof) can be written recursively. If it has more than a single leaf, the density is the density of the first branching time $\delta_{\langle 0\rangle}$ times the density of the two descendant subtrees. If there is only a single leaf the density is just the probability of that species actually being extinct at time t.

If $T^o(t) = (\tau^o(t), \{x_\iota\}_{\iota \in V(\tau^o(t))})$ is considered a realization of $\mathscr{T}_r(t)$,

$$f(T^o_\iota(t)) = \begin{cases} d^t(x_\iota, x_{m\iota}) f(T^o_{\iota\langle 1\rangle}(t)) f(T^o_{\iota\langle 2\rangle}(t)), & \text{if} \quad |\breve{V}(T^o_\iota)| > 1 \\ p_1(t - x_{m\iota}), & \text{if} \quad |\breve{V}(T^o_\iota)| = 1 \end{cases} \tag{10}$$

where $T^o_{\iota\langle 1\rangle}, T^o_{\iota\langle 2\rangle}$ are the descendant subtrees of $\iota \in \mathring{V}(T^o(t))$. The statement follows by recursively applying the first case of eq. (10) until each subtree has only one leaf, i.e. those rooted at internal nodes of $T^o(t)$, then applying the second case for the leaf nodes. □

Again, we are not actually interested in oriented trees, but probability densities for the underlying unoriented time tree, and for time-phylogenies with the same tree shape. These are easily found by a multiplicative constant of $2^{n-1-c(\tau(t))}$ for the unoriented time tree and $2^{n-1}/n!$ for the time-phylogeny respectively. Note that we count cherries instead of symmetry vertices in the former constant as we can always distinguish between subtrees of a vertex when considering their branch lengths too, as these are almost surely different. If we furthermore ignore the branch lengths of the unoriented time-tree, we get a tree shape, whose density (now actually just a probability, as the tree shape space is discrete) can be found by marginalizing over the branching times for the internal nodes,

Theorem 4.4. *Let $\tau(t)$ be the (unoriented) tree shape of the reconstructed tree $\mathscr{T}_r(t)$ at a time $t \geq 0$. The density of each re-rooted subtree $\tau_\iota(t)$ for $\iota \in \mathring{V}(\tau_\iota(t))$ given their origin time, that is their parent's branching time, satisfies the following recursion*

$$f(\tau_\iota(t) \mid x_{m\iota}) = \begin{cases} \nu(\tau_\iota(t)) \int_{x_{m\iota}}^t d^t(x_\iota, x_{m\iota}) f(\tau_{\iota\langle 1\rangle}(t) \mid x_\iota) f(\tau_{\iota\langle 2\rangle}(t) \mid x_\iota)\, \mathrm{d}x_\iota, & |\breve{V}(\tau_\iota)| > 1 \\ p_1(t - x_{m\iota}), & |\breve{V}(\tau_\iota)| = 1 \end{cases}$$

under the condition that $x_{m\langle 0\rangle} = 0$ and where $\nu(\tau_\iota(t))$ is 2 when the descendant subtree shapes of ι are equal, i.e. if ι is a symmetry vertex in $\tau(t)$.

Proof. When the subtree τ_ι has more than one leaf we simply marginalize the joint density eq. (10) over the possible values of the branching time, this must be later than the branching point of the parent of ι but no later than t. $\nu(\tau_\iota(t))$ takes care of the oriented trees which have the same tree shape. If there is only one leaf on the subtree is just the probability that that branch actually stays a branch until t. □

Note that with the condition that $x_{m\langle 0\rangle} = 0$, which by definition is true, this recursion actually provides the marginal distribution $f(\tau(t))$ for the

tree shape, an expression containing $n-1$ integral, many of which are nested. If we further condition the tree on having n leaves, the density is simply found by normalizing $f(\tau(t))$ for $\tau(t)$ with n leaves with the total probability of tree shapes with n leaves,

$$f(\tau(t) \mid n) = \frac{f(\tau(t))}{\sum_{\gamma \in \mathrm{TS(n)}} f(\gamma(t))}$$

where $\mathrm{TS(n)}$ is the set of tree shapes with n leaves and $f(\gamma(t))$ is the density of such a tree shape under the binary Bellman-Harris process sampled at time t.

We can interpret all the densities here as conditional densities given that the process was sampled at time t after its beginning, but in practice, we seldom know this age of the process as we don't know the time at which the MRCA actually became a species, that is time 0 is not really known. One way to deal with this uncertainty is to put a prior distribution on t, for example, a uniform distribution $f(t \mid t_{\max})$ over $(0, t_{\max})$ for some maximal possible age $t_{\max}$ of the process. Conditional on the number of leaves on the tree this prior is

$$\begin{aligned} f(t \mid n, t_{\max}) &= \frac{f(n \mid t) f(t \mid t_{\max})}{f(n)} \\ &= \frac{f(n \mid t)/t_{\max}}{\int_0^{t_{\max}} f(n \mid u)/t_{\max} \,\mathrm{d}u} \\ &= \frac{\sum_{\gamma \in \mathrm{TS(n)}} f(\gamma(t))}{\int_0^{t_{\max}} \sum_{\gamma \in \mathrm{TS(n)}} f(\gamma(u)) \,\mathrm{d}u} \end{aligned}$$

This does not really solve the problem, as we now just have to know some $t_{\max}$, but we could let $t_{\max} \to \infty$, hence getting an improper prior,[5]

$$f(t \mid n) = \frac{\sum_{\gamma \in \mathrm{TS(n)}} f(\gamma(t))}{\int_0^{\infty} \sum_{\gamma \in \mathrm{TS(n)}} f(\gamma(u)) \,\mathrm{d}u}$$

under the assumption that the denominator is finite. Gernhard, Hartmann, and M. Steel (2008, Theorem 4) has shown that for the binary Bellman-Harris process without extinctions ($q = 1$), the denominator is finite when the lifetime distribution has finite mean. The proof does not rely on the fact that there are no extinctions in the process, so with a small adaptation, the proof holds for the process with extinctions as well.

Using this prior, we can get a time-independent tree shape density under the binary Bellman-Harris process conditioned on having n leaves,

$$f(\tau \mid n) = \int_0^{\infty} f(\tau(t) \mid n) f(t \mid n) \,\mathrm{d}t = \frac{\int_0^{\infty} f(\tau(t)) \,\mathrm{d}t}{\int_0^{\infty} \sum_{\gamma \in \mathrm{TS(n)}} f(\gamma(u)) \,\mathrm{d}u}. \tag{11}$$

The results of this section are, when taking into account the translation between oriented trees and tree shapes, equivalent to those in Jones (2011a) for theorem 4.3 and Gernhard, Hartmann, and M. Steel (2008) and Hagen et al. (2015) for theorem 4.4.

[5] This approach is usual for phylogenetic models (D. Aldous and Popovic 2005; Gernhard, Hartmann, and M. Steel 2008).

4.4 Tree balance

With the tree shape $\tau(t)$ we could consider the balance of the tree as described in section 3.4, but now we would of course achieve a splitting distribution dependent on time as well as the descendant subtree sizes. Knowing the functions $P_n(t)$ we would get the time-dependent splitting probabilities as the probability of a $j : n-j$ split normalized by the probability of any $k : n-k$ split for $k \in [n-1]$,

$$s_n(j,t) = \frac{p_j(t)p_{n-j}(t)}{\sum_{k=1}^{n-1} p_k(t)p_{n-k}(t)}, \quad \text{for } j \in [n-1],\, t \geq 0$$

and again we could assign a uniform prior over $(0,\infty)$ to t and get a time-independent splitting distribution

$$s_n(j) = \frac{\int_0^\infty p_j(t)p_{n-j}(t)\,\mathrm{d}t}{\sum_{k=1}^{n-1}\int_0^\infty p_k(t)p_{n-k}(t)\,\mathrm{d}t}$$

as argued in Jones (2011a). It is demonstrated in Jones (2011b, Figure 3) and Hagen et al. (2015) that the binary Bellman-Harris process can produce trees with a balance close to that of empirical trees with a properly chosen lifetime distribution and speciation probability.

4.5 The Yule process

The *Yule process* (Yule 1925) is arguably the simplest continuous-time model for tree evolution, allowing only speciations and no extinctions, hence the process is a *pure-birth process*, but we will see that the process is a special case of the binary Bellman-Harris process.

Standard expositions of Yule processes e.g. (Karlin and Taylor 1975, Chap. 4.1) consider it as a Markov process $X := \{X(t)\}_{t\geq 0}$ with state space $\mathbb{N}_0$ counting the number of species. Note that the background space on which this process lives can be chosen as an arbitrarily enlargened version of $(\Omega, \mathbb{A}, \mathbb{P})$ to make X a measurable mapping on it, while Z also remains a measurable mapping as discussed. The number of species can increase in unit steps with transition rate $N\lambda$ when the number of species is currently N. The process cannot decrease. This corresponds to assuming each individual species speciates independently with a rate λ and never goes extinct, but we do not keep track of which species are speciating at each event. For a given $t \geq 0$ we let $P_n(t) := \mathbb{P}(X(t) = n)$ for $n \geq 1$, these probabilities must satisfy the *forward differential equations*

$$P_n'(t) = -n\lambda P_n(t) + (n-1)\lambda P_{n-1}(t), \quad \text{for } n \geq 1$$

which under the condition $P_1(0) = 1$, i.e. that there are only one species at time zero, has the solution

$$P_n(t) = e^{-\lambda t}(1-e^{-\lambda t})^{n-1}, \quad \text{for } n \geq 1, \tag{12}$$

and $X(t)$ then has generating function,

$$F_X(s,t) := \sum_{n=1}^{\infty} P_n(t)s^n = \frac{e^{-\lambda t}s}{1-(1-e^{-\lambda t})s}, \quad \text{for } s \in \overline{D}(0;1). \tag{13}$$

The Yule branching process

Let us instead consider a Bellman-Harris process Z with lifetime distribution $G(t) = 1 - e^{-\lambda t}$, i.e. an exponential distribution with rate λ, and let $q = 1$, such that we have $h(s) = s^2$. These choices correspond to speciations happening independently within a lineage according to an exponential distribution with rate λ and that there are no extinctions in the process, just like in our Markov process X. We will now see that $Z \overset{\mathcal{D}}{=} X$.

By T. E. Harris (1963, Theorem VI 11.1), the generating function of Z satisfies the differential equation as its lifetime distribution is exponential

$$\frac{\partial F_Z(s,t)}{\partial t} = \lambda\Big(h\big(F_Z(s,t)\big) - F_Z(s,t)\Big) = \lambda\Big(F_Z(s,t)^2 - F_Z(s,t)\Big). \tag{14}$$

Taking the t-derivative of F_X (eq. (13)) shows that is a solution to eq. (14), and since the solution to eq. (14) is unique (T. E. Harris 1963, Theorem VI 9.4), we must have that $F_Z = F_X$. Generating functions determine the underlying distributions completely, so X and Z are equally distributed, in particular, is Z also a Markov process.

This shows that the Yule model can be considered a special case of the Bellman-Harris branching process. So while the Markov process X itself does not contain any genealogical information about the species it counts, we can just consider the branching tree of Z as the underlying tree of X.

The Yule tree

As Z and X have the same distribution, we have from eq. (12) that $p_n(t) = e^{-\lambda t}(1-e^{-\lambda t})^{n-1}$ for $n \geq 1$, $t \geq 0$, and since we have no extinctions, $p_0(t) = 0$ for all $t \geq 0$. In particular, we note that $p_1(t) = e^{-\lambda t}$.

Now the density of the reconstructed MRCA (which is now simply the MRCA as the reconstructed tree is equal to the complete tree) becomes $d^t(s,u) = g(s-u) = \lambda e^{-\lambda(s-u)}$ which is independent of the time since the origin of the process, hence we drop the superscript t. We then get the density of the oriented time tree $T^o(t)$ being

$$\begin{aligned} f(T^o(t)) &= \prod_{\iota \in \mathring{V}(T^o(t))} \lambda e^{-\lambda(x_\iota - x_{m\iota})} \prod_{\iota \in \breve{V}(T^o(t))} e^{-\lambda(t - x_{m\iota})} \\ &= \lambda^{n-1} \prod_{\iota \in V(T^o(t))} e^{-\lambda(x_\iota^t - x_{m\iota})} \end{aligned}$$

where $x_\iota^t = t$ for leaf vertices. Moreover Gernhard, Hartmann, and M. Steel (2008) shows that the tree shape density $f(\tau(t))$ for the Yule process is

$$f(\tau(t)) = 2^{n-1-s(\tau)} \frac{e^{-\lambda t}(1-e^{-\lambda t})^{n-1}}{\prod_{v \in \mathring{V}(T)} \ell_v - 1}$$

where $\ell_v = \left|\{u \mid u \in \breve{V}(T), v \prec u\}\right|$ is the number of descendant leaves of v, and $s(\tau)$ is the number of symmetry nodes in τ.

If we consider splitting distributions of the Yule tree, it turns out that they are independent of the time since the origin of the process, and actually equal to the splitting distribution of the Yule-Harding model,

$$s_j(n,t) = \frac{p_j(t)p_{n-j}(t)}{\sum_{k=1}^{n-1} p_k(t)p_{n-k}(t)} \tag{15}$$

$$= \frac{e^{-\lambda t}(1-e^{-\lambda t})^{j-1}e^{-\lambda t}(1-e^{-\lambda t})^{n-j-1}}{\sum_{k=1}^{n-1} e^{-\lambda t}(1-e^{-\lambda t})^{k-1}e^{-\lambda t}(1-e^{-\lambda t})^{n-k-1}} \tag{16}$$

$$= \frac{e^{-2\lambda t}(1-e^{-\lambda t})^{n-2}}{\sum_{k=1}^{n-1} e^{-2\lambda t}(1-e^{-\lambda t})^{n-2}} \tag{17}$$

$$= \frac{1}{n-1} \tag{18}$$

More generally, D. Aldous (1996) argues that $f(\tau \mid n)$ of eq. (11) (multiplied by $2^{s(\tau)}/n!$ to make it a distribution over $\mathrm{RB}(n)$, not over tree shapes) is exactly the Yule-Harding distribution (proposition 3.2) under the Yule process.

Birth-Death processes

The Yule process is as mentioned a pure-birth process. If we allow for extinctions in our branching process we get a *(constant rate) birth-death process*, which like the Yule process is usually assumed to be Markovian[6], and is presented in a traditional Markov process framework in most textbooks (Karlin and Taylor 1975). But like the Yule process, the birth-death process is also a special case of our binary Bellman-Harris branching process, and analytic solutions for $p_n(t)$ and $d^t(s,u)$ for the birth-death process are presented in Jones (2011a, Theorem 3).

Even though this process contains extinctions, the last result of the last section, that $f(\tau \mid n)$ is the Yule-Harding distribution under the Yule process, holds even for more general birth-death processes. D. Aldous (1996) argues that any process ending with n extant species and which satisfy the following produces phylogenies with the Yule-Harding distribution:

> *"At each time t there are a finite number of species alive (starting with one species at time 0). From time to time there is an "event" which is either an extinction or a speciation, i.e. either some species B becomes extinct or some species A splits into species A and A'. The time from t until the next event, and the chance the next event is an extinction rather than a speciation, may depend on the past in an arbitrary way. But if the next event is an extinction then each species is equally likely to be the one to become extinct, and if the next event is an speciation then each species is equally likely to be the one to speciate."*

[6]Age-dependent birth-death processes have been studied (Kendall 1949), but if not directly stated we usually assume they are Markovian.

The only non-trivial component of this condition for birth-death processes is that extinctions and speciations are equally likely to occur in each extant species at time t, but this is a consequence of the lifetimes being exponentially distributed. Consider the extant individuals at some time $t \geq 0$, $I_e(t, \omega) = \{\iota \in I(\omega) \mid \sigma_\iota \leq t < \delta_\iota\}$. Each $\iota \in I_e(t)$ has been extant for a time u_ι such that $\sigma_\iota + u_\iota = t$. Now the probability that any $\iota \in I_e(t)$ speciates or goes extinct before time $t + s$ for some $s \geq 0$ given that they are extant at time t becomes

$$\mathbb{P}(L_\iota \geq u_\iota + s \mid L_\iota \geq u_\iota) = \mathbb{P}(L_\iota \geq s) \tag{19}$$

by the memorylessness of the exponential distribution. This probability is the same for all extant species, so there must be an equal probability for each species to speciate or go extinct first.

This shows that birth-death processes generate Yule-Harding distributed phylogenies. Is this the case for all binary Bellman-Harris processes? The answer is no, as the argument we gave above strongly relies on the memorylessness of the lifetime distribution, which is only a property of the exponential and the geometric distribution. Actually, Hagen et al. (2015) shows that under a suitably chosen lifetime distribution, the binary Bellman-Harris process without extinctions can produce trees with a balance on the spectrum between the Yule-Harding model and the uniform model.

5 Discussion

In this report, we have reviewed elements of phylogenetic tree evolution going from simple discrete tree models to a general branching process model for modeling macroevolution as happening through time under a set of quite general assumptions.

The methods presented go from the very earliest work in this field, as done by Yule (1925) and Harding (1971) to the initial work on phylogenetic time-tree in A W F Edwards (1970) and Z. Yang and Rannala (1997) and to the modern work including Jones (2011a), Gernhard, Hartmann, and M. Steel (2008), and Stadler (2010).

5.1 Applications

Our presentation have solely been of a theoretical nature and have not touched upon use cases of the theory. We will now consider this briefly. The general model of Jones (2011a) and Hagen et al. (2015) has not seen many applications yet, probably due to their mathematical difficulty and novelty, hence the birth-death model and extensions of it are still the workhorse of phylogenetic analyses. We will describe how the general model could be worthwhile to work with for future analyses.

Testing evolutionary hypotheses

Ecologists could be interested in assessing the effect of speciations and extinctions through time, that is macroevolution, on the diversity of species, possibly with an interest in whether the observed data can be attributed to random forces alone, or whether there have been some biological forces affecting the outcome. To test such hypotheses, a null-model is required, i.e. a model under which we can predict evolution as happened due to simple stochastic laws. In the case of a good fit between data and the null-model, this model might prove a good approximation to reality, while a bad fit suggests that there exists strong biological forces not considered in the model (Hey 1992).

To compare the predictions of a stochastic model to data we would like to estimate the parameters of the model. For example, in a constant-rate birth-death model, one would estimate the speciation and extinction rates, which could be done using maximum likelihood theory (Ziheng Yang, Goldman, and Friday 1995). Under the fitted model we could compare predictions on phylogenetically important questions, such as tree balance, and as mentioned throughout the report we generally see a discrepancy between tree balance in the simple models and in empirical trees (Blum and François 2006). This suggests that the models do not capture the actual speciation and extinction dynamics, a problem that more general models, such as the binary Bellman-Harris model, might remedy (Hagen et al. 2015). While our binary Bellman-Harris model has shown to fit empirical trees well under a suitable lifetime distribution and speciation probability, further generalizations and different directions of study have been considered and might be worthwhile to develop further.

While touching upon estimation in these models it should be noted that there has recently been a surge in interest in whether the parameters of the more complex models are even identifiable. For example, it has been shown that for a constant-rate birth-death model with random sampling of the extant species is not identifiable unless we know the rate of sampling (Stadler 2009). Furthermore under the generalized birth-death model (Kendall 1948) with time-varying speciation and extinctions rates, it has been shown that these rate functions are in general not identifiable (Louca and Pennell 2020). Legried and Terhorst (2023) however has shown that by restricting the class of functions, we can still identify the rate functions.

Bayesian tree inference

Reconstructing the phylogeny given a set of species has traditionally been solely a task of finding the best tree in some sense given the observed extant species. Methods such as maximum parsimony favor the tree which involved "the minimum net amount of evolution", while distance-based methods are essentially clustering methods that recursively cluster species that are in some sense evolutionarily close. But uncertainty measures of the reconstructed trees are not considered (Felsenstein 2004).

In contrast, we could consider inferring the phylogeny given some data a Bayesian problem, loosely stated we have

$$\mathbb{P}(T \mid D) = \frac{\mathbb{P}(D \mid T)\mathbb{P}(T)}{\mathbb{P}(D)}$$

where T is a phylogeny and D is the observed data, for example, the set of extant species. If we could actually find the posterior distribution $\mathbb{P}(T \mid D)$ then we could both obtain a reconstructed tree (e.g. via maximum a posteriori estimation) and it becomes a trivial task to compare the probability of the reconstructed tree with other possible trees. But a key component to finding the posterior distribution is to choose a prior distribution over the phylogenies.

The models presented in this report all give rise to a prior distribution over phylogenies. The uniform model and birth-death models are often used as the prior model for tree evolution in practical analyses with two different motivations. The uniform model is used as an uninformative prior, that is we would like to influence the Bayesian analysis as little as possible with former knowledge, hence we just choose to give equal probability to all phylogenies (Alfaro and Holder 2006). On the other hand with the birth-death model, we try to give a likely process of evolution as we think it might occur (Z. Yang and Rannala 1997). As we learn more and more about the evolution of species, a Bayesian would argue that this prior knowledge should be accounted for in future analyses, hence a more general model for the prior distribution might be useful in future analyses. Several properties of the binary Bellman-Harris model make it attractable to use as a prior distribution, see Jones (2011a).

5.2 Generalizations and limitations

While the binary Bellman-Harris process might give rise to more precise phylogenetic analyses, there are, as there always will be with any simplistic model of a complex system, some important limitations to its use.

First and foremost the models we have presented here do not necessarily capture how we sample our data. One aspect is that we assume that all species which are generated from the ancestor are sampled at time t which is rarely the case. In practice, one could possibly sample some proportion of the extant species at random, and this should be taken into account in the model. However, if we assume that our sampling is random at time t, the model proposed in Jones (2011a) does take this into account, we did not include it to keep the presentation relatively concise. See also Stadler (2009) for more on sampling in birth-death models. Our data could moreover contain non-extant species as well, this has not been investigated for the binary Bellman-Harris process but has been considered for constant-rate birth-death models (Stadler 2010).

Further generalizations of the evolutionary dynamics could also be considered. While the presented model already relaxes some of the assumptions from the regularly used models, there are still important characteristics of evolution that are ignored. We have assumed throughout that in

a speciation event the ancestral species go extinct and two new species come into existence, but this might not be the case as offspring species and the ancestral species might go on to exist simultaneously, and the ancestral species should not be given an age of 0 at the speciation point then. For Markovian models, this is no problem due to memorylessness, but in age-dependent models, the age of the ancestral species should be carried over even though it has an offspring species. Hagen et al. (2015) suggest "asymmetric speciation" in the binary Bellman-Harris process taking this into account, but while this works in simulated models, a Bellman-Harris process is not the right tool if we want to investigate this analytically as it always assumes that the ages of both species are 0 right after speciation (T. E. Harris 1963). Meanwhile, this could be treated using a Crump-Mode-Jagers process (Jagers 1975), which might be an interesting further development.

Likewise, the lifetime distribution and speciation probability, and hence the speciation and extinction rates, might be extended to be time-varying. As mentioned this has been done for the birth-death model as used in phylogenetics (Rabosky 2006), but for more general branching process models this has not been considered in a phylogenetic context. Kimmel (1983) and more recently Pakkanen et al. (2023) have considered such time-varying branching processes from a theoretical point of view and in infectious disease epidemiology respectively.

A practical limitation is the problem of simulating from the model. Although simulating conditional only on the time since the origin of the process is straightforward, we know of no more efficient simulation method for the binary Bellman-Harris model than directly sampling lifetimes and speciation probabilities one at a time and keeping track of each individual species at all times. For Markovian models, more efficient simulation procedures exist such as the Gillespie algorithm (Gillespie 2007). More problematic is the situation where we want to condition on having n species at time t or simply conditioning on having n leaf species[7], as it would possibly require a huge number of samples using the direct sampling methods. More efficient schemes have been proposed for birth-death models (Hartmann, Wong, and Stadler 2010; Höhna 2013) and for the wider class of *coalescent point process models* (Lambert and Stadler 2013).

Lastly, evolutionary processes such as *lateral gene transfer* and *incomplete lineage sorting* might make the whole phylogenetic tree framework too simple. For ILS we might want to consider extensions where the gene tree is allowed to evolve stochastically "inside" the random species tree (Mossel and Roch 2010), while LTG might need models evolving as a more general network to be fully understood (M. A. Steel 2016).

[7]Actually conditioning on the phylogeny having n species is ambiguous when dealing with models with extinctions. There might be several time-intervals in which the model has n extant species (Hartmann, Wong, and Stadler 2010)

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